ROAD OBJECT DETECTION FISHEYE CAMERA

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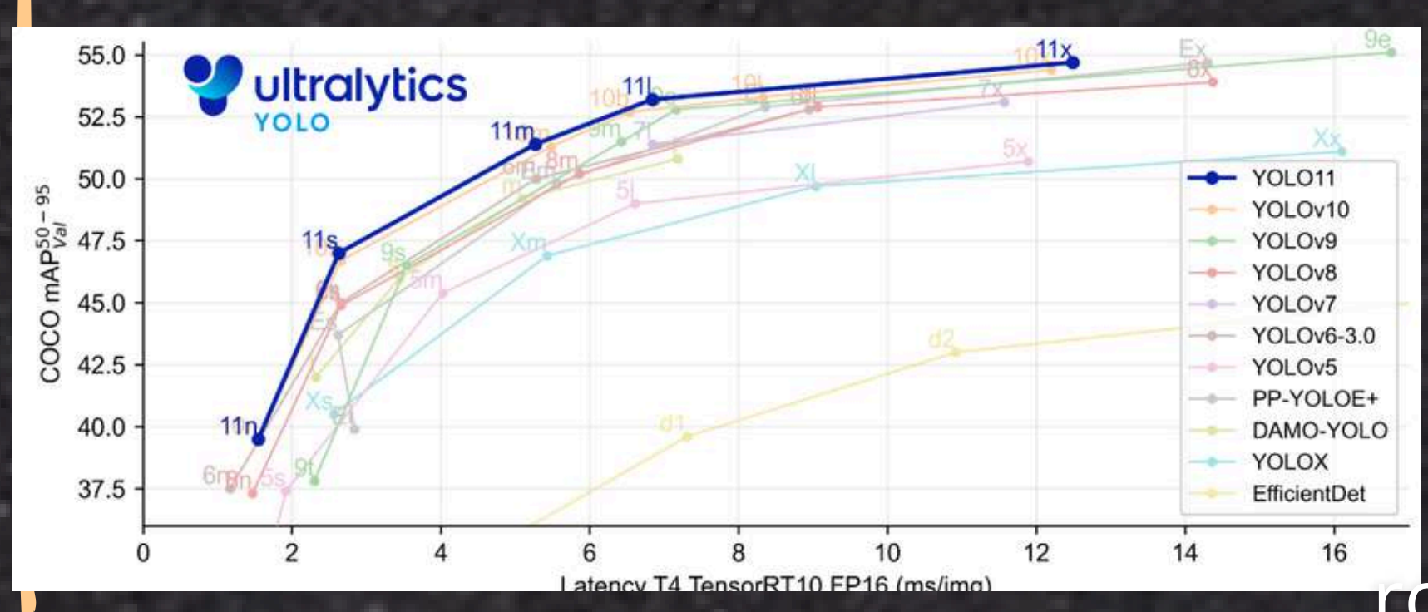
Introduction

Road object detection is crucial for autonomous driving, intelligent transportation, and traffic management. By accurately identifying vehicles, pedestrians, cyclists, and traffic signs, these systems improve safety and navigation. For example, real-time detection enables autonomous vehicles to make rapid, informed decisions, reducing accident risk. Traffic management also benefits, using detection to monitor road use, manage congestion, and optimize flow, supporting smarter cities.

Objective

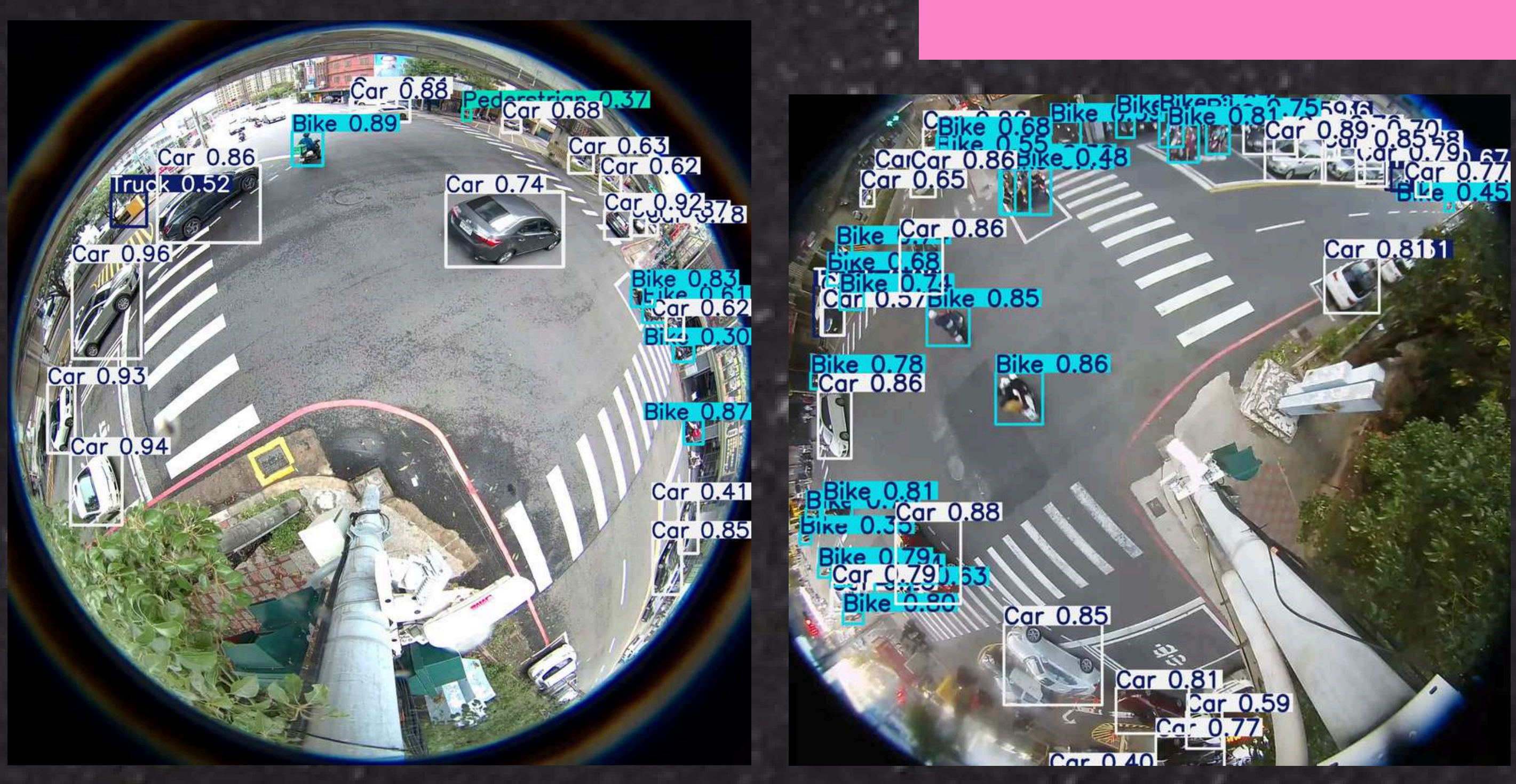


This research aims to compare the performance of YOLOv9, YOLOv10, and YOLO11 for detecting road objects using the Fisheye8K dataset. Key performance metrics—such as accuracy, speed, and handling of fisheye image distortion—will be evaluated to determine which model is best suited for real-time road object detection. This study seeks to inform future advancements in autonomous driving and traffic management by identifying the optimal YOLO version for these applications.

Methodology

Overall, larger models tend to be more effective for the Fisheye dataset, as demonstrated by the superior performance of YOLOv11x and YOLOv9e in terms of mAP, F1-score, and Recall. These models, which are designed to handle more complex feature extraction and capture finer details, show a clear advantage in detecting objects with high precision across various object scales. The increased model capacity allows them to adapt to the unique characteristics of the Fisheye dataset, which includes challenges such as distortion and varied object sizes. Smaller models like YOLOv10s and YOLO11s struggled with these complexities, exhibiting lower performance metrics. Thus, for tasks requiring high accuracy and robustness in object detection, larger models are better suited to handle the diverse nature of the Fisheye dataset.

Result



Predicted model using YOLO11

Conclusion

Our evaluation results show that YOLOv9e and YOLO11x models , which are pretrained on MS-COCO, outperforms the other models. Therefore the FishEye8K dataset will be a significant contribution to the fisheye video analytics and smart city applications. Future work includes further benchmarking the YOLO model on additional datasets to evaluate its performance across a wider range of street object categories and scenarios.